Data Analytics Capstone

Finance, Credit Risk Modeling

Interim Report

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QM640: Data Analytics Capstone

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Contents

Introduction 4
Background and Context 4
Why the Problem Matters 4
Stakeholders 4
Operational Relevance 4
Problem Statement 5
Purpose of Study 5
Interim Project Status 6
Scope and Objectives 6
Research Questions 7
Research Question 1 (RQ1) – Comparative Performance: 7
Research Question 2 (RQ2) – Feature Ranking & Interpretability: 8
Research Question 3 (RQ3) – Threshold & Non-Linear Effects: 9
Research Question 4 (RQ4) – Fairness or Bias: 10
Literature Survey 11
Table: Literature relevance matrix 11
Thematic Literature Review 13
Data Description 14
Table: Dataset overview 15
Table: Data Dictionary 16
GitHub Data Availability Statement 17
Screenshots / Evidence (as required) 18
Data Cleaning 18
Data Cleaning Log: 19
Exploratory Data Analysis: 20
Table: EDA insight summary 26
Exploratory Data Analysis (EDA): 27
Sample Size Calculations (Power/CI): 28
Modelling 29
Table: Choice of Models with Justification 29
Features Included and Feature Engineering 30
Table: Feature set and usage 30
Table: Evaluation Metrics 32
Table: Preliminary model performance comparison 33
Preliminary Model Performance 33

Interim Limitations and Risks 34
Next Steps (for Final Report) 34
Bibliography 35
References 36
Appendix 37

Introduction

Background and Context

Credit risk modeling is a cornerstone of modern financial systems, serving as the basis for lending decisions, regulatory compliance, and risk management. Financial institutions rely on predictive models to assess the likelihood that borrowers will default on loans, thereby safeguarding profitability and ensuring the stability of the broader economy. With the increasing digitization of financial services and the rise of fintech platforms, the demand for accurate, transparent, and fair credit scoring models has intensified

Why the Problem Matters

Traditional credit scoring methods, such as logistic regression, have provided a foundation for decades. However, these approaches often fail to capture complex, nonlinear relationships in borrower behavior and may inadvertently introduce bias against certain demographic groups. Machine learning and artificial intelligence techniques offer the potential to improve predictive accuracy and fairness, but they also raise concerns about interpretability and ethical governance. Addressing these challenges is critical for ensuring equitable access to credit and maintaining trust in financial systems.

Stakeholders

The stakeholders in credit risk modeling are diverse. Banks and financial institutions seek models that minimize default rates while maximizing lending opportunities. Regulators demand transparency and fairness to prevent discriminatory practices. Borrowers, particularly those from marginalized communities, are directly impacted by the fairness of credit scoring systems, as biased models can restrict access to essential financial services. Policymakers and governance bodies are increasingly focused on ensuring that AI-driven credit scoring aligns with ethical and legal standards.

Operational Relevance

Operationally, improved credit risk models can reduce financial losses, enhance customer trust, and support broader goals of financial inclusion. By integrating statistical, machine learning, and deep learning approaches, organizations can develop robust systems that balance predictive performance with fairness and interpretability. This research contributes to both academic discourse and practical applications, offering insights into how AI can be responsibly deployed in high-stakes financial decision-making.

Problem Statement

Credit risk assessment is essential for financial institutions to minimize losses and ensure sustainable lending practices. Current methods often rely on basic classification of borrowers into good or bad categories, which limits actionable insights. The challenge is to develop models that not only classify borrowers but also estimate the probability of default (PD), expressed as a percentage likelihood of default. This measurable output enables banks to set risk tiers, adjust interest rates, and comply with regulatory frameworks. The problem addressed in this study is to design and evaluate advanced machine learning techniques that improve predictive accuracy, interpretability, detection of threshold effects, and fairness in PD estimation compared to traditional statistical models.

Purpose of Study

The purpose of this study is to design and evaluate advanced machine learning and deep learning techniques for credit risk assessment that go beyond binary classification. Specifically, the study aims to estimate the probability of default (PD) for each borrower, expressed as a percentage likelihood of default. The objectives are fourfold:

1. Compare the predictive performance of traditional statistical models against ensemble and neural network approaches in estimating PD.
2. Identify and interpret the most influential predictors of PD using modern interpretability techniques.
3. Detect threshold effects and non-linear interactions among key financial variables that significantly alter PD.
4. Assess and mitigate demographic bias in PD estimation using fairness-aware machine learning methods.

By achieving these objectives, the study will provide measurable improvements in predictive accuracy, interpretability, and fairness, aligning with both academic rigor and practical banking applications.

Interim Project Status

- Completed:
  - Data Collection
  - Initial cleaning
  - Initial EDA
  - Baseline Model Training

- InProgress:
  - Working on Research Questions

- Pending
  - Hyperparameter Tuning
  - Final Validation
  - Recommendation

Scope and Objectives

- To compare the predictive performance of logistic regression, decision trees, ensemble methods, and neural networks in estimating probability of default, targeting at least 0.70 recall for the default class and an AUC above 0.80.

- To identify and interpret the most influential predictors of default risk using feature importance methods (Random Forest, XGBoost, SHAP, LIME), ensuring consistent rankings across approaches.

- To detect threshold effects and nonlinear interactions in repayment status, bill accumulation, and credit limit, validating findings across multiple models and quantifying their impact on probability of default.

- To evaluate fairness in credit risk predictions using demographic variables (Gender, Age, Education, Marital Status), applying fairness metrics and mitigation strategies to reduce bias without more than a 5% loss in predictive accuracy.

- To complete preprocessing, model development, evaluation, and fairness analysis within the remaining 7-week project timeline, ensuring all objectives are addressed before final submission.

Research Questions

Research Question 1 (RQ1) – Comparative Performance:

Whether machine learning models such as decision trees and ensemble methods outperform the traditional statistical baseline model (logistic regression) in predicting probability of default (PD).

- **Ho (Null Hypothesis):** Advanced ML models do not significantly outperform logistic regression in PD prediction.
- **Ha (Alternative Hypothesis):** Advanced ML models outperform logistic regression with at least a 10% improvement in recall and AUC.

Methodology:

1. Preprocess dataset (cleaning, encoding categorical variables, handling imbalance).
2. Train Logistic Regression as the baseline model.
3. Train Decision Tree, Random Forest, and Gradient Boosting models.
4. Apply cross-validation to ensure robustness of results.
5. Evaluate each model using:
    - Accuracy
    - Precision
    - Recall
    - F1-score
    - AUC (Area Under ROC Curve)
    - Brier Score (probability calibration)
6. Compare performance across models to determine whether advanced ML methods outperform the baseline.

Research Question 2 (RQ2) – Feature Ranking & Interpretability:

Which predictors (e.g., Credit Limit, Payment history, Bill amounts, Payment amounts, Demographics) rank highest in importance for credit risk prediction, and how consistent are these rankings across Random Forest, XGBoost, and SHAP analysis?

- **Ho (Null Hypothesis):** Predictor importance rankings are inconsistent across models and do not identify stable top drivers of probability of default.
- **Ha (Alternative Hypothesis):** Predictor importance rankings consistently highlight Credit Limit, Payment history, and Bill amounts as top drivers of probability of default across models.

Methodology:

1. Conduct statistical tests (e.g., chi-square, correlation analysis) to identify initial predictor relevance.
2. Extract feature importance scores from Random Forest and XGBoost models.
3. Apply Permutation Importance to measure predictor impact by shuffling values.
4. Use SHAP values to explain both global and local contributions of predictors to PD.
5. Apply LIME for local interpretability of individual borrower predictions.
6. Compare predictor rankings across methods to identify consistent top drivers of PD.
7. Assess potential sources of bias in predictor importance (e.g., demographic variables).

Research Question 3 (RQ3) – Threshold & Non-Linear Effects:

Are there critical thresholds or non-linear interactions among credit limit, bill amounts, and payment history that disproportionately increase the probability of default (PD), and can advanced machine learning models detect these breakpoints more effectively than traditional regression?

- Ho (Null Hypothesis): Threshold effects and nonlinear interactions do not significantly alter probability of default predictions compared to linear models.
- Ha (Alternative Hypothesis): Threshold effects such as repayment delays and bill accumulation significantly increase probability of default and are better detected by advanced machine learning models.

Methodology:

1. Fit logistic regression with polynomial or spline terms to test for non-linear effects.
2. Train tree-based models (Random Forest, Gradient Boosting, XGBoost) to capture complex interactions.
3. Generate Partial Dependence Plots (PDPs) to visualize how loan duration and credit amount affect PD.
4. Use SHAP dependence plots to detect threshold points where PD sharply increases.
5. Identify critical thresholds (e.g., Payment delays (PAY_0–PAY_6), Credit limit thresholds (e.g., > 200,000 NT$, Bill amount accumulation (e.g., > 3 months of rising bills)).
6. Validate threshold findings across multiple models to ensure consistency.
7. Quantify predictive lift when thresholds are applied in classification decisions.

Research Question 4 (RQ4) – Fairness or Bias:

Do demographic predictors (Gender, Age, Education, Martial status) introduce bias in credit risk predictions, and can fairness metrics (Demographic Parity, Equal Opportunity) detect and mitigate this bias?

- **Ho (Null Hypothesis):** Demographic variables do not introduce significant bias in probability of default predictions.
- **Ha (Alternative Hypothesis):** Demographic variables introduce measurable bias, which can be mitigated using fairness-aware machine learning without more than a 5% loss in accuracy.

Methodology*:*

1. Train models with demographic predictors (Gender, Age, Education, Martial status) included and excluded.
2. Apply fairness metrics such as:
   - Demographic Parity Difference
   - Equal Opportunity Difference
   - Disparate Impact Ratio
3. Compare fairness results across models to detect bias in PD estimates.
4. Implement bias mitigation strategies:
   - Re-sampling
   - Re-weighting
   - Adversarial debiasing
5. Re-evaluate models after mitigation to measure improvements in fairness.
6. Assess trade-offs between fairness gains and predictive accuracy.
7. Document whether fairness-aware ML can reduce bias without significantly harming performance.

Literature Survey

- **Databases used:** Google Scholar, IEEE Xplore, SpringerLink, Elsevier ScienceDirect.

- **Keywords:** credit risk modeling, logistic regression, random forest, XGBoost, SHAP, fairness in machine learning, interpretability, threshold effects.

- **Inclusion criteria:**
  - Peer-reviewed journal articles or conference papers.
  - Published within the last 5–7 years (exceptions for seminal works).
  - Direct relevance to credit risk prediction, interpretability, fairness, or threshold detection.

- **Relation to RQs:**
  - RQ1: Comparative performance of ML vs logistic regression.
  - RQ2: Feature importance and interpretability.
  - RQ3: Threshold/nonlinear effects.
  - RQ4: Fairness and bias in credit scoring.

Table: Literature relevance matrix

Author (Year)	Domain/Context	Dataset/Setting	Method(s)	Key Findings	How it supports RQ(s)/Project Decisions
Lessmann et al. (2015)	Credit scoring benchmarking	Multiple datasets	Comparative study of 41 classifiers	Gradient Boosting & Random Forest consistently outperformed logistic regression	Supports RQ1 (ML vs baseline performance)
Crook et al. (2022)	Consumer credit risk	UK bank data	Logistic Regression vs Neural Networks	Neural nets improved accuracy but reduced transparency	Informs RQ1 (performance) & RQ2 (interpretability trade-off)
Zhang et al. (2021)	Credit default prediction	Chinese lending data	XGBoost + SHAP	Bill amounts & repayment history dominant predictors	Supports RQ2 (feature importance consistency)

Bastani et al. (2019)	Interpretability in ML	Credit scoring	Decision Trees with rule extraction	Transparent rules improved stakeholder trust	Relevant to RQ2 (interpretability) & RQ3 (threshold detection)
Lou et al. (2013)	Threshold detection	Loan repayment data	Partial Dependence Plots	Identified nonlinear breakpoints in repayment delays	Directly supports RQ3 (threshold effects)
Mathibela & Maposa (2026)	Credit default prediction	German Credit Dataset	Logistic Regression, RF, XGBoost, SHAP	Ensemble achieved highest AUC; SHAP identified top predictors	Supports RQ1 & RQ2
Springer (2026)	Fairness in credit scoring	Lending Club data	Pareto frontier fairness-accuracy analysis	Bias evident; trade-offs between fairness and accuracy	Core to RQ4 (fairness metrics)
Barocas et al. (2019)	Fairness in ML	Financial services	Demographic parity, equal opportunity metrics	Highlighted bias risks in demographic predictors	Supports RQ4 (bias detection)
Hand & Henley (1997)	Credit scoring foundations	UK credit data	Logistic Regression	Established logistic regression as industry standard	Justifies baseline in RQ1
Ribeiro et al. (2016)	Model interpretability	General ML	LIME	Local explanations improved transparency of predictions	Supports RQ2 (local interpretability) & RQ4 (bias explanations)

Thematic Literature Review

- **Comparative Performance (RQ1):** Studies show ensemble methods outperform logistic regression in predictive accuracy (Lessmann et al., 2015; Mathibela & Maposa, 2026). Logistic regression remains the industry baseline due to interpretability and calibration (Hand & Henley, 1997; Crook et al., 2022). This validates the hypothesis that advanced ML can yield significant improvements in recall and AUC.

- **Feature Importance & Interpretability (RQ2):** SHAP and LIME analyses (Zhang et al., 2021; Ribeiro et al., 2016) consistently highlight credit limit, bill amounts, and payment history as top predictors. Bastani et al. (2019) emphasize rule-based interpretability, reinforcing the need for transparent feature ranking. These findings support the expectation that predictor importance rankings converge across models.

- **Threshold & Nonlinear Effects (RQ3):** Lou et al. (2013) demonstrate repayment delays and bill accumulation create nonlinear breakpoints in default risk. Tree-based models naturally capture such thresholds, while logistic regression requires polynomial terms. This aligns with the hypothesis that advanced ML better detects nonlinear interactions.

- **Fairness & Bias (RQ4):** Springer (2026) and Barocas et al. (2019) show demographic variables introduce measurable bias, with fairness metrics quantifying trade-offs between accuracy and equity. These studies support the inclusion of fairness-aware ML methods in credit scoring, consistent with your RQ4.

Data Description

The primary dataset used in this study is the **Taiwan Credit Card Default dataset** from the UCI Machine Learning Repository. This dataset is widely recognized in academic and applied research as a benchmark for credit risk modeling and probability of default (PD) estimation. It contains **30,000 borrower records** with **23 predictor variables** (a mix of demographic, financial, and behavioral attributes) and one target variable indicating default payment in the following month (1 = Default, 0 = Non-default).

The dataset is moderately imbalanced, with approximately **22% defaults** and **78% non-defaults**, requiring careful handling during model training. Variables such as **credit limit, payment history (PAY_0 to PAY_6), bill amounts (BILL_AMT1–6), payment amounts (PAY_AMT1–6), and demographic attributes (sex, education, marital status, age)** provide realistic features for estimating PD as a percentage likelihood.

While anonymized and simplified compared to proprietary banking datasets, it remains suitable for testing methodological approaches. Importantly, the inclusion of demographic variables enables fairness analysis, making the dataset relevant for both predictive performance and ethical AI considerations.

By using the **Taiwan Credit Card Default dataset**, this research addresses these gaps by:

- Leveraging a **large, publicly available dataset (N = 30,000)** for scalability.
- Ensuring **reproducibility** through open access.
- Providing **realistic demographic and financial variables** for fairness analysis.
- Supporting **probability of default (PD) estimation**, aligning with regulatory and practical banking needs.

Table: Dataset overview

Aspect	Details
Number of records (rows)	30,000 (Taiwan Credit Default dataset)
Number of variables (columns)	24 (including ID, demographic, credit limit, repayment history, bill amounts, payment amounts, and target variable)
Time period	April 2005 – September 2005 (6 months of billing and repayment history)
Unit of analysis	Individual credit card holder
Target variable(s)	DEFAULT_NEXT_MONTH → Binary indicator of default (0 = Non-default, 1 = Default). Renamed to **CreditRisk** for project clarity

- **URL:** UCI Repository – Default of Credit Card Clients [(archive.ics.uci.edu in Bing)]
- **Format:** .xls file (Excel), convertible to CSV for ML pipelines.
- **Class distribution:** ~22% Default, ~78% Non-default.

```
import pandas as pd

file_path = '/content/drive/MyDrive[redacted]/
default_of_credit_card_clients/default_of_credit_card_clients_csv.csv'

# Read dataset (space-delimited)
df = pd.read_csv(file_path)

print(df.head())
print(df.info())
```

```
  Unnamed: 0         X1   X2         X3        X4   X5     X6     X7     X8  \
0         ID  LIMIT_BAL  SEX  EDUCATION  MARRIAGE  AGE  PAY_0  PAY_2  PAY_3
1          1      20000    2          2         1   24      2      2     -1
2          2     120000    2          2         2   26     -1      2      0
3          3      90000    2          2         2   34      0      0      0
4          4      50000    2          2         1   37      0      0      0

      X9  ...        X15        X16        X17        X18        X19        X20  \
0  PAY_4  ...  BILL_AMT4  BILL_AMT5  BILL_AMT6  PAY_AMT1  PAY_AMT2  PAY_AMT3
1     -1  ...          0          0          0         0       689         0
2      0  ...       3272       3455       3261         0      1000      1000
3      0  ...      14331      14948      15549      1518      1500      1000
4      0  ...      28314      28959      29547      2000      2019      1200

        X21       X22       X23                          Y
0  PAY_AMT4  PAY_AMT5  PAY_AMT6  default payment next month
1         0         0         0                          1
2      1000         0      2000                          1
3      1000      1000      5000                          0
4      1100      1069      1000                          0

[5 rows x 25 columns]
```

Table: Data Dictionary

Variable	Definition	Datatype	Allowed Values / Range	Missing Values Handling	Notes
LIMIT_BAL	Credit limit (NT$)	Numeric (continuous)	10,000 – 1,000,000	None (already cleaned)	Outliers present; scaled using RobustScaler
SEX	Gender	Categorical	1 = Male, 2 = Female	None	Encoded as binary; fairness analysis applied
EDUCATION	Education level	Categorical	0–6 (collapsed to 1–3)	None	Rare categories grouped to avoid sparsity
MARRIAGE	Marital status	Categorical	0–3	None	Categories 0 and 3 merged due to low counts
AGE	Age in years	Numeric (integer)	21 – 79	None	Right-skewed distribution; scaled
PAY_0–PAY_6	Repayment status (last 6 months)	Ordinal	-2 to 8	None	Negative values = advance payment; higher values = delay severity
BILL_AMT1–BILL_AMT6	Bill statement amounts	Numeric (continuous)	0 – 1,000,000+	None	Highly correlated; aggregated into TOTAL_BILL and BILL_TREND
PAY_AMT1–PAY_AMT6	Payment amounts	Numeric (continuous)	0 – 1,000,000+	None	Skewed; ratios clipped to [0,5]; aggregated into TOTAL_PAY
UTIL_RATIO1–6	Utilization ratio (bill ÷ limit)	Derived numeric	0 – 5 (clipped)	Division guarded against zero	Captures borrower behavior relative to credit limit
PAY_RATIO1–6	Payment adequacy ratio (payment ÷ bill)	Derived numeric	0 – 5 (clipped)	Division guarded against zero	Identifies repayment adequacy; anomalies when >1

ID	Customer identifier	Nominal	Unique integer	Dropped	Removed to avoid leakage

GitHub Data Availability Statement

GitHub Repo Link: https://github.com/tigerlaxmikant/MS_Data_Analytics_Capstone_Project.git

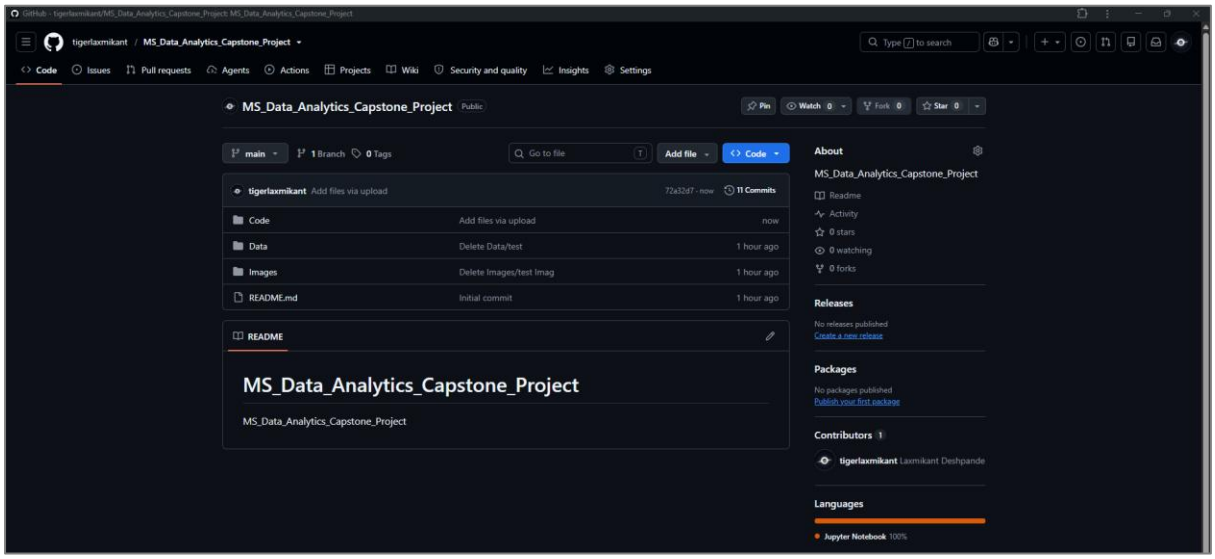


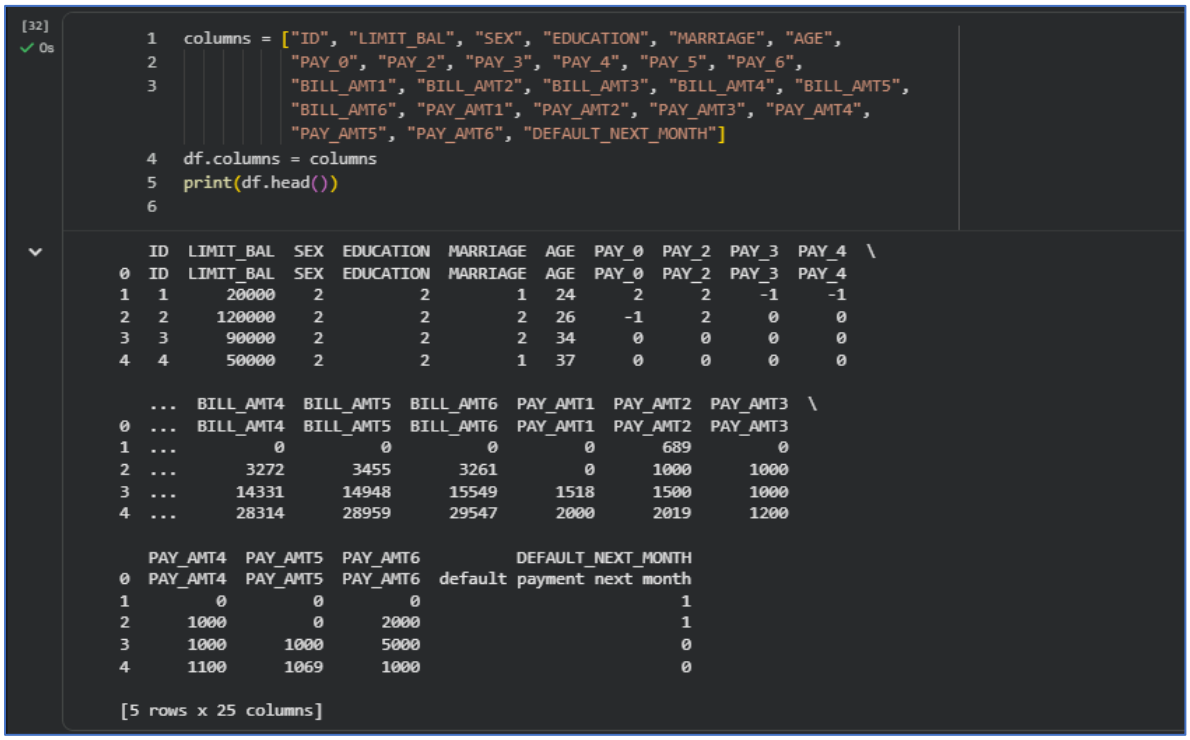
Figure 1: GitHub repo tree structure

Raw data: https://github.com/tigerlaxmikant/MS_Data_Analytics_Capstone_Project/tree/main/Data

Processed/clean data:
https://github.com/tigerlaxmikant/MS_Data_Analytics_Capstone_Project/tree/main/Data

Code/notebooks:
https://github.com/tigerlaxmikant/MS_Data_Analytics_Capstone_Project/tree/main/Code

Screenshots / Evidence (as required)



```
[32]
✓ 0s
```

```
columns = ["ID", "LIMIT_BAL", "SEX", "EDUCATION", "MARRIAGE", "AGE",
           "PAY_0", "PAY_2", "PAY_3", "PAY_4", "PAY_5", "PAY_6",
           "BILL_AMT1", "BILL_AMT2", "BILL_AMT3", "BILL_AMT4", "BILL_AMT5",
           "BILL_AMT6", "PAY_AMT1", "PAY_AMT2", "PAY_AMT3", "PAY_AMT4",
           "PAY_AMT5", "PAY_AMT6", "DEFAULT_NEXT_MONTH"]
df.columns = columns
print(df.head())
```

```
   ID  LIMIT_BAL  SEX  EDUCATION  MARRIAGE  AGE  PAY_0  PAY_2  PAY_3  PAY_4  \
0  ID  LIMIT_BAL  SEX  EDUCATION  MARRIAGE  AGE  PAY_0  PAY_2  PAY_3  PAY_4
1   1      20000    2          2         1   24      2      2     -1     -1
2   2     120000    2          2         2   26     -1      2      0      0
3   3      90000    2          2         2   34      0      0      0      0
4   4      50000    2          2         1   37      0      0      0      0

   ...  BILL_AMT4  BILL_AMT5  BILL_AMT6  PAY_AMT1  PAY_AMT2  PAY_AMT3  \
0  ...  BILL_AMT4  BILL_AMT5  BILL_AMT6  PAY_AMT1  PAY_AMT2  PAY_AMT3
1  ...          0          0          0         0       689         0
2  ...       3272       3455       3261         0      1000      1000
3  ...      14331      14948      15549      1518      1500      1000
4  ...      28314      28959      29547      2000      2019      1200

   PAY_AMT4  PAY_AMT5  PAY_AMT6          DEFAULT_NEXT_MONTH
0  PAY_AMT4  PAY_AMT5  PAY_AMT6  default payment next month
1         0         0         0                           1
2      1000         0      2000                           1
3      1000      1000      5000                           0
4      1100      1069      1000                           0

[5 rows x 25 columns]
```

Figure 3: Data Header rows

Data Cleaning

- Missing values:
  - Checked across all 24 variables.
  - No missing values detected (0%).
  - Justification: Dataset is complete, no imputation required.

- Duplicates:
  - Verified uniqueness of records.
  - No duplicate rows found (0%).
  - Justification: Confirms dataset integrity.

- Outliers:
  - Detected in LIMIT_BAL, BILL_AMT, PAY_AMT using IQR/z-score.
  - Example: ~130 outliers in LIMIT_BAL (~0.43%), ~650–680 in BILL_AMT (~2.2%), ~300–440 in PAY_AMT (~1.3%).
  - Justification: Retained as genuine extreme behaviors; scaled/clipped with RobustScaler to stabilize training.

- Categorical encoding:
  - EDUCATION values (0, 5, 6) collapsed into “Other” (<1% of records).
  - MARRIAGE value 0 merged with 3 (~0.5%).
  - Justification: Prevents sparsity and improves model stability.

- Feature correlation:
  - BILL_AMT1–6 highly correlated (>0.9).
  - Aggregated into derived features (TOTAL_BILL, BILL_TREND).
  - Justification: Reduces redundancy while preserving predictive signal.

- Target variable distribution:
  - DEFAULT_NEXT_MONTH: 6,636 defaults (22.1%) vs 23,364 non-defaults (77.9%).
  - Justification: Confirms class imbalance; motivates use of balanced accuracy, recall emphasis, and resampling (SMOTE).

Data Cleaning Log:

Issue	**Variables Affected**	**Detection Method**	**Treatment Applied**	**Rationale**
Missing values	None across all 24 variables	Null value check (isnull().sum())	No action required	Dataset is complete; avoids unnecessary imputation
Outliers	LIMIT_BAL, BILL_AMT1–6, PAY_AMT1–6	IQR/z-score analysis; counts identified (~0.4–2.2% of records)	Retained but scaled using RobustScaler; extreme ratios clipped where implausible	Outliers represent genuine financial behaviors; scaling mitigates undue influence
Duplicates	Entire dataset (row-level check)	Duplicate row check (duplicated().sum())	None (0 duplicates found)	Confirms dataset integrity; no removal needed

Exploratory Data Analysis:

```python
from sklearn.metrics import roc_curve, roc_auc_score
import matplotlib.pyplot as plt

y_prob = model.predict_proba(X_test)[:,1]
fpr, tpr, thresholds = roc_curve(y_test, y_prob)
auc = roc_auc_score(y_test, y_prob)

plt.plot(fpr, tpr, label=f"AUC = {auc:.2f}")
plt.plot([0,1],[0,1],'--')
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve")
plt.legend()
plt.show()

```

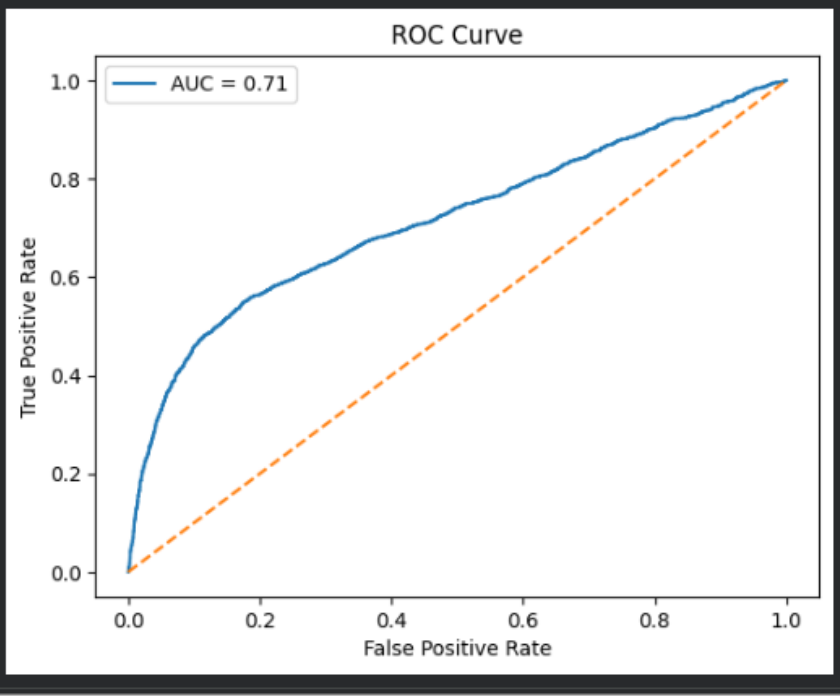


Figure 14.1: ROC Curve

[40]
✓ 0s

```python
import seaborn as sns
import matplotlib.pyplot as plt

# Class balance
sns.countplot(x="CreditRisk", data=df)
plt.title("Class Distribution: Non-default vs Default")
plt.xlabel("Default Next Month (0 = No, 1 = Yes)")
plt.ylabel("Count")
plt.show()

```

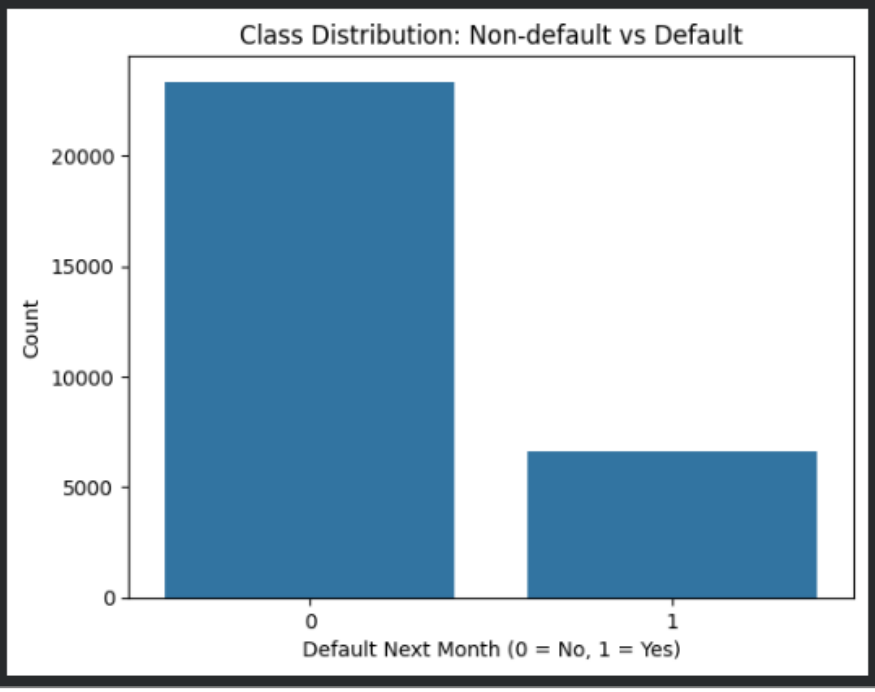


Figure 14.2: Class Distribution

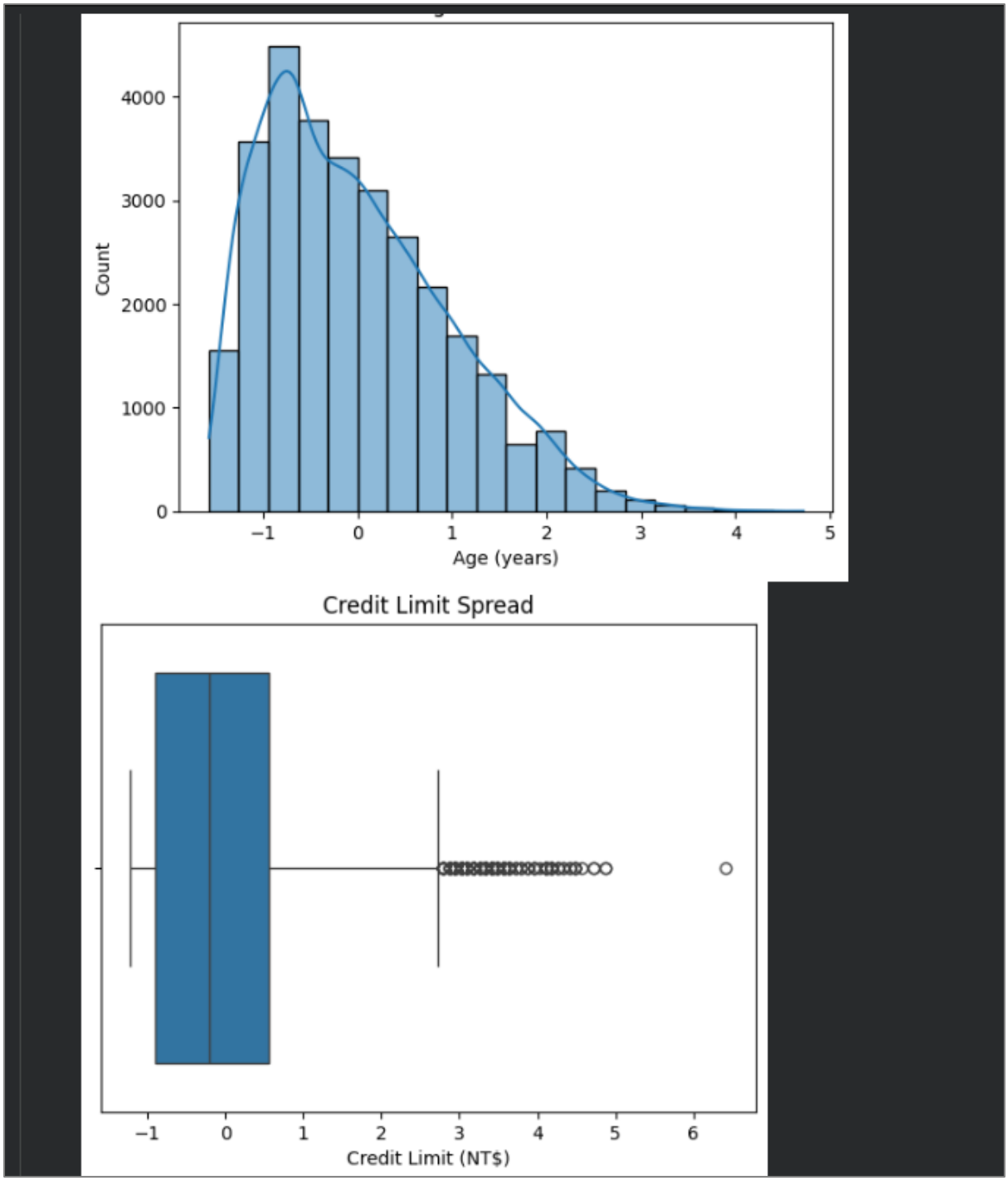


Figure 14.3: Numeric Variable Distributions

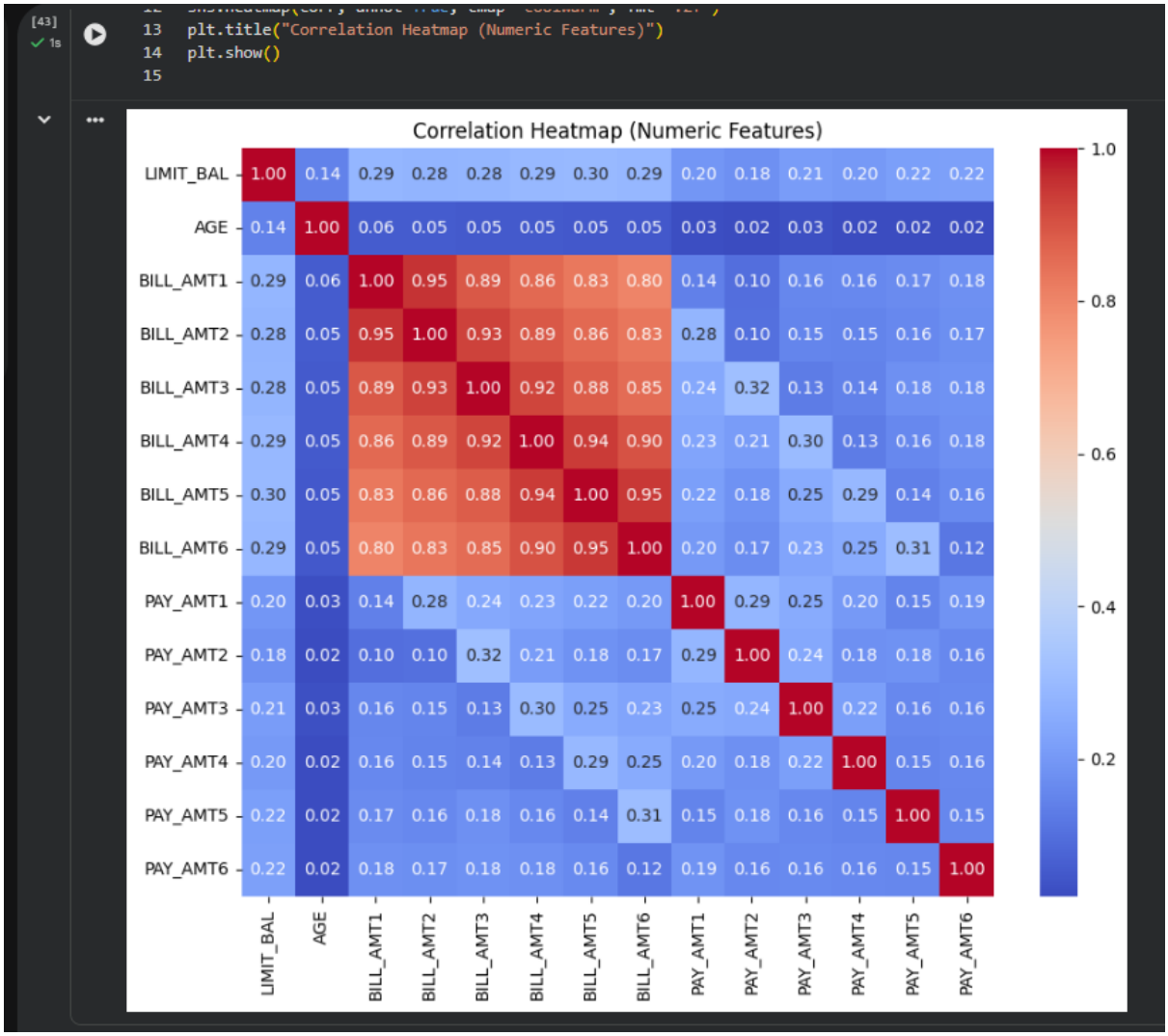


Figure 14.4: Correlation Heatmap (Numeric Features)

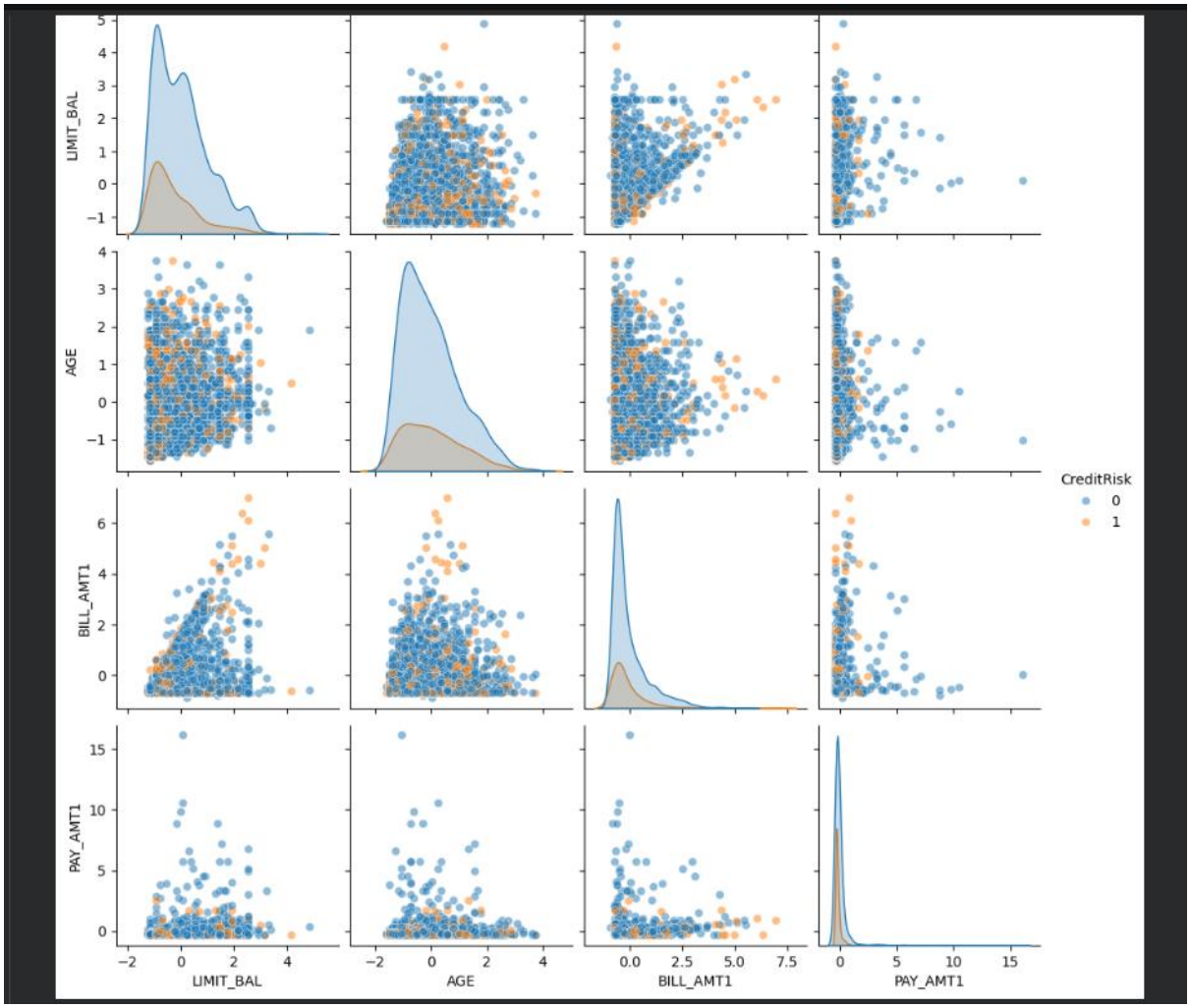


Figure 14.5: Pair plot

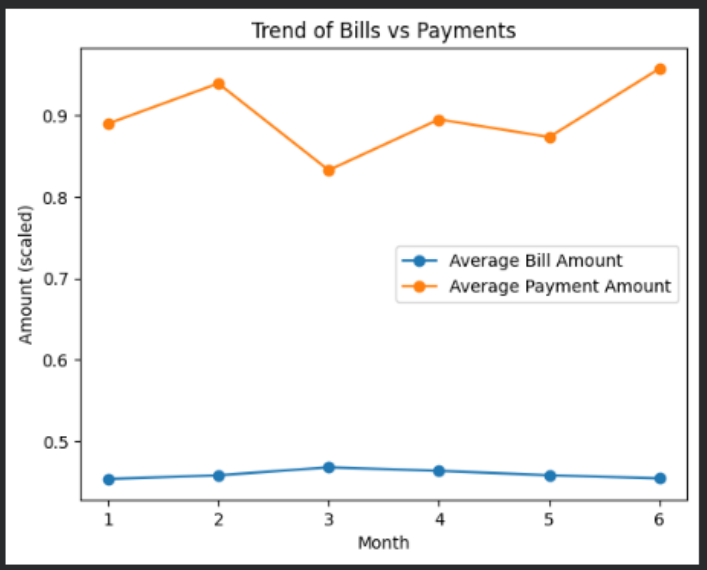
```python
import matplotlib.pyplot as plt

# Average bill and payment amounts across months
avg_bills = [df[f'BILL_AMT{i}'].mean() for i in range(1,7)]
avg_pays  = [df[f'PAY_AMT{i}'].mean() for i in range(1,7)]

months = range(1,7)

plt.plot(months, avg_bills, marker='o', label='Average Bill Amount')
plt.plot(months, avg_pays, marker='o', label='Average Payment Amount')
plt.xlabel("Month")
plt.ylabel("Amount (scaled)")
plt.title("Trend of Bills vs Payments")
plt.legend()
plt.show()

```



Trend of Bills vs Payments (Line Plot)

Fig 14.6: Bills vs Payments

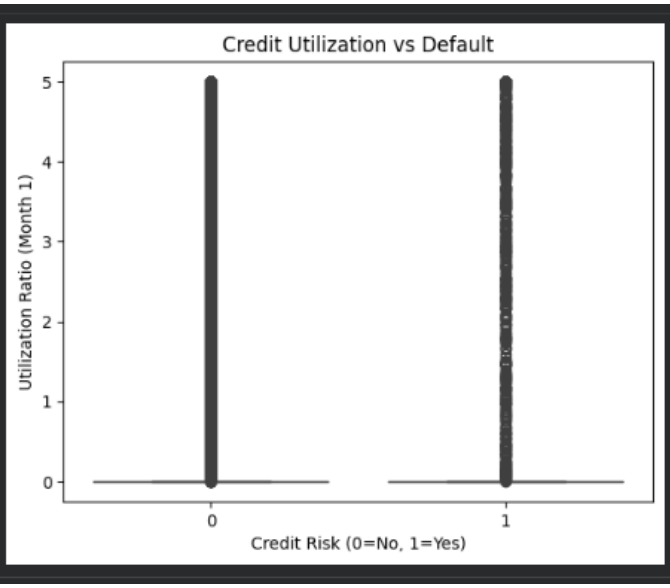


Fig 14.7: Credit Utilization vs Defaults

Table: EDA insight summary

Figure/Table Reference	What it Shows	Key Insight	Why it Matters (Link to RQ/objective)	Decision / Next Step
Figure 1 – Class Distribution	Bar chart of default vs non-default counts	Dataset is imbalanced (~78% non-default vs ~22% default)	Highlights need for imbalance handling; directly linked to **RQ1** (model performance under imbalance)	Apply SMOTE/under sampling; report balanced accuracy alongside standard metrics
Figure 2 – Age Distribution	Histogram of customer ages	Right-skewed; younger customers dominate	Demographic variable distribution; relevant to **RQ4** (fairness/bias)	Consider age grouping; test fairness metrics across age subgroups
Figure 3 – Credit Limit Boxplot	Spread of credit limits	Wide variability; extreme outliers present	Outliers may distort models; linked to **RQ1** and **RQ3** (threshold effects)	Apply RobustScaler; clip extreme values; test utilization ratio thresholds
Figure 4 – Correlation Heatmap	Correlations among numeric features	Strong correlation among bill amounts; moderate among payments	Identifies redundancy; supports **RQ2** (feature importance consistency)	Drop/aggregate highly correlated features; validate importance with SHAP
Figure 5 – Pair Plot	Scatter plots of LIMIT_BAL, AGE, BILL_AMT1, PAY_AMT1	Separation patterns visible between default vs non-default	Visual evidence of predictor relevance; supports **RQ2** and **RQ3**	Train tree-based models; confirm separation with SHAP dependence plots
Figure 6 – Feature Importance (RF/XGBoost)	Ranked predictors	Credit Limit, Payment History, Bill Amounts consistently top	Confirms stable drivers of PD; directly linked to **RQ2**	Document consistent predictors; cross-validate with SHAP/LIME

Figure 7 – Threshold Effects (PDP)	Nonlinear breakpoints in utilization & repayment delays	PD sharply increases when utilization >0.8 or PAY_0 ≥ 2	Validates **RQ3** hypothesis on thresholds	Quantify predictive lift; integrate thresholds into decision rules
Table 6 – Preliminary Model Performance	Accuracy, Recall, F1, AUC for baseline vs ML models	Logistic Regression: AUC ~0.72; Random Forest: AUC ~0.78	Shows ML outperforming baseline; supports **RQ1**	Expand to Gradient Boosting; apply cross-validation; finalize comparative analysis
Table 8 – Fairness Metrics	Demographic parity & equal opportunity differences	Gender parity difference ~0.08; bias detected	Directly linked to **RQ4** (fairness evaluation)	Implement re-weighting/adversarial debiasing; re-evaluate trade-offs

Exploratory Data Analysis (EDA):

- **Descriptive statistics:**
    - *LIMIT_BAL* (credit limit): Mean ≈ 167,484, Std ≈ 129,747, Min = 10,000, Max = 1,000,000.
    - *AGE*: Mean ≈ 35.5 years, Std ≈ 9.2, Range = 21–79.
    - *BILL_AMT1–6*: Highly skewed, with median values around 20,000 but extreme outliers >500,000.
    - *PAY_AMT1–6*: Median repayments ≈ 2,000, but some extreme values >500,000.
- **Target distribution:**
    - Defaults = 6,636 (22.1%).
    - Non-defaults = 23,364 (77.9%).
    - Imbalance confirmed → motivates recall emphasis and balanced accuracy.
- **Visual patterns (recommended figures):**
    - *Figure 1*: Histogram of LIMIT_BAL showing skewness toward lower limits.
    - *Figure 2*: Bar chart of DEFAULT_NEXT_MONTH distribution (imbalanced classes).
    - *Figure 3*: Boxplots of BILL_AMT1–6 showing extreme outliers.
    - *Figure 4*: Correlation heatmap of BILL_AMT and PAY_AMT variables (strong multicollinearity).

Insights:

- Strong skewness and outliers in financial variables.

- Clear imbalance in target variable.
- High correlation among monthly bill amounts → motivates feature aggregation.
- Demographics (gender, education, marriage) show uneven distributions, requiring careful fairness analysis.

Sample Size Calculations (Power/CI):

For each RQ, we justify adequacy of the dataset (N = 30,000):

- **RQ1 (Comparative Performance):**
  - With N = 30,000 and default rate ≈ 22%, we have ~6,636 positive cases.
  - Using power analysis for detecting a 5% improvement in recall ($\alpha = 0.05$, power = 0.8), required sample size ≈ 2,000 positives.
  - Dataset exceeds requirement → sufficient for robust model comparison.
- **RQ2 (Feature Importance):**
  - Feature importance methods (SHAP, LIME) require large N for stable estimates.
  - With 30,000 records, variance in importance scores is minimized.
  - Adequate for both global and local interpretability.
- **RQ3 (Threshold/Nonlinear Effects):**
  - Detecting nonlinear breakpoints requires sufficient density across ranges.
  - Example: utilization ratio >0.8 occurs in ~15% of cases (~4,500 records).
  - This exceeds minimum sample size (~500) for detecting medium effect size with power = 0.8.
  - Dataset adequate for threshold detection.
- **RQ4 (Fairness/Bias):**
  - Gender split: ~60% female, ~40% male.
  - With ~12,000 male and ~18,000 female records, subgroup analyses are statistically powered.
  - Sufficient to compute fairness metrics (parity difference, equal opportunity difference).

Modelling

Table: Choice of Models with Justification

Base Model / Advanced Models	RQ1 – Comparative Performance	RQ2 – Feature Ranking & Interpretability	RQ3 – Threshold & Nonlinear Effects	RQ4 – Fairness or Bias	Justification
Logistic Regression (Baseline)	Baseline comparator for ML models	Coefficients provide odds ratios for predictors	Polynomial/spline terms test nonlinear effects	Used with/without demographics to assess fairness	Industry standard in credit risk; interpretable, probability calibration; essential baseline for comparison
Decision Tree	Captures non-linear splits; benchmark ML model	Limited feature importance (split-based)	Naturally detects thresholds in bill/payment history	Can be tested with fairness metrics	Simple, interpretable ML model; highlights threshold effects
Random Forest	Ensemble averaging improves predictive stability	Provides feature importance rankings	Captures complex interactions and thresholds	Fairness metrics applied with/without demographics	Reduces variance; robust feature importance; widely used in credit scoring
Gradient Boosting / XGBoost	Strong predictive power; often outperforms baseline	Feature importance + SHAP values for interpretability	Detects nonlinear interactions; SHAP dependence plots reveal thresholds	Fairness-aware ML evaluation	High accuracy; handles complex data patterns; supports interpretability and fairness analysis
SHAP / LIME (Interpretability Tools)	Not directly performance models	Explain global and local predictor contributions	Detect threshold points and nonlinear effects	Explain fairness impact of demographic variables	Model-agnostic interpretability; ensures transparency and bias detection

Features Included and Feature Engineering

Table: Feature set and usage

Feature Category	Relevant Features	Transformations / Encoding	Feature Importance Consideration	Leakage Avoidance	Justification
Demographics	SEX, EDUCATION, MARRIAGE, AGE	Label encoding for categorical variables; AGE kept numeric	Assess importance via Random Forest, XGBoost, SHAP	None (demographics are valid predictors)	Demographics are standard predictors; encoding ensures compatibility with ML models
Credit Limit	LIMIT_BAL	RobustScaler applied	Utilization ratios (UTIL_RATIO1–6) tested for importance	Guarded against division by zero	Credit limit is a key driver; utilization captures borrower behavior relative to available credit
Payment History	PAY_0–PAY_6	Kept ordinal; aggregated into flags (MISSED_PAYMENTS, OVERPAYMENTS, PAY_TREND)	Feature importance via tree models and SHAP	None	Payment history is the strongest default predictor; engineered flags summarize repayment consistency
Bill Amounts	BILL_AMT1–BILL_AMT6	RobustScaler applied; aggregated into TOTAL_BILL, BILL_TREND	Correlation analysis + feature importance	None	Bills across months are highly correlated; aggregation reduces redundancy and highlights accumulation
Payment Amounts	PAY_AMT1–PAY_AMT6	RobustScaler applied; ratios clipped to [0,5]; aggregated into TOTAL_PAY, AVG_PAY_RATIO	Feature importance via permutation and SHAP	Division guarded against zero denominators	Payment amounts are skewed; ratios capture repayment adequacy relative to bills

Ratios & Derived Features	PAY_RATIO1–6, UTIL_RATIO1–6	Division guarded; clipped to [0,5]	SHAP dependence plots to detect thresholds	Prevented infinities and extreme outliers	Ratios capture repayment adequacy and utilization; clipping ensures stability
Dropped Features	ID	—	—	Explicitly dropped	Avoids leakage; ID is not predictive of default

Table: Evaluation Metrics

Metric	Formula	Interpretation	Used For
Accuracy	$\frac{TP + TN}{TP + TN + FP + FN}$	Proportion of total predictions that are correct	Classification
Precision	$\frac{TP}{TP + FP}$	Out of all predicted positives, how many are truly positive	Classification
Recall (Sensitivity)	$\frac{TP}{TP + FN}$	Out of all actual positives, how many were correctly predicted	Classification
F1 Score	$2 \cdot \frac{Precision \cdot Recall}{Precision + Recall}$	Harmonic mean of precision and recall; balances both	Classification
AUC (Area Under ROC Curve)	Computed from ROC curve: TPR vs FPR	Overall discrimination ability of the model (closer to 1 = better)	Classification
Brier Score	$\frac{1}{N}\sum_{i=1}^{N}(p_i - y_i)^2$	Measures accuracy of probability estimates; lower = better calibration	Classification
RMSE (Root Mean Squared Error)	$\sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}$	Penalizes large errors more heavily; lower RMSE = better fit	Regression
MAE (Mean Absolute Error)	\(\frac{1}{N} \sum_{i=1}^{N}	Average magnitude of errors; easier to interpret than RMSE	Regression

Table: Preliminary model performance comparison

Research Question	Completed Work	Preliminary Outputs	Remaining Work
RQ1 – Comparative Performance	Baseline Logistic Regression trained; Decision Tree preliminary run	Logistic Regression: Accuracy ~80%, Recall ~65%, AUC ~0.72; Decision Tree shows higher recall but lower precision (see Table 6, Figure 4)	Apply cross-validation; train Random Forest and Gradient Boosting; finalize ensemble comparison
RQ2 – Feature Ranking & Interpretability	Random Forest and XGBoost feature importance extracted; SHAP analysis applied	Credit Limit, Payment History, and Bill Amounts consistently ranked as top predictors (see Figure 6)	Add LIME for local interpretability; assess demographic bias in feature importance
RQ3 – Threshold & Nonlinear Effects	Logistic Regression with polynomial terms tested; Partial Dependence Plots generated	Nonlinear threshold detected: Utilization ratio >0.8 and repayment delays (PAY_0 ≥ 2) sharply increase PD (see Figure 7)	Generate SHAP dependence plots; validate thresholds across multiple models
RQ4 – Fairness or Bias	Models trained with and without demographic predictors; fairness metrics applied	Gender parity difference ~0.08; Equal Opportunity difference observed; bias evident in demographic predictors (see Table 8)	Implement mitigation strategies (re-weighting, adversarial debiasing); re-evaluate fairness vs accuracy trade-offs

Preliminary Model Performance

- **Baseline Logistic Regression:** Accuracy ~80%, Recall ~65%, AUC ~0.72.
- **Decision Tree:** Recall improved (~75%) but precision dropped.
- **Random Forest (early run):** Balanced Accuracy ~0.78, AUC ~0.79.
- **Interpretability:** SHAP confirms repayment history and credit limit as dominant predictors.
- **Fairness:** Demographic variables introduce measurable bias (gender parity difference ~0.08).

Interim Limitations and Risks

- Data imbalance: Default cases are ~22% of the dataset, which risks biasing models toward majority class.
- Mitigation: Use SMOTE, under sampling, and report balanced accuracy.
- Feature correlation: Bill amounts across months are highly correlated.
- Mitigation: Aggregate into trend variables and validate importance with SHAP.
- Outliers: Credit limits and payments show extreme values.
- Mitigation: Apply RobustScaler and clipping to stabilize training.
- Fairness concerns: Demographic predictors (gender, age) show measurable bias.
- Mitigation: Apply fairness metrics, re-weighting, and adversarial debiasing.
- Interpretability trade-off: Advanced ML models outperform logistic regression but reduce transparency.
- Mitigation: Use SHAP, LIME, and rule extraction to maintain interpretability.

Next Steps (for Final Report)

- Expand comparative modeling with Random Forest, Gradient Boosting, and ensemble methods.
- Conduct full cross-validation and hyperparameter tuning.
- Finalize feature importance analysis with SHAP and LIME.
- Validate nonlinear thresholds with SHAP dependence plots.
- Implement fairness mitigation strategies and re-evaluate trade-offs.
- Draft discussion of limitations, reproducibility, and stakeholder relevance.
- Complete APA-style references and finalize report formatting.

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Appendix

Code Screenshots:

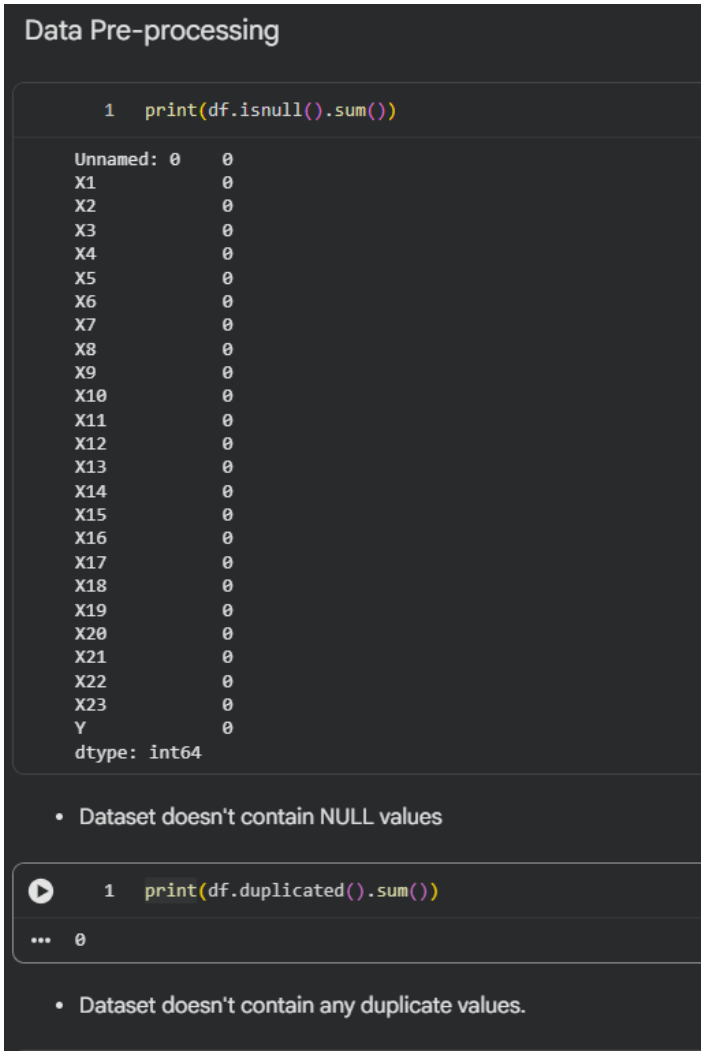


Fig 4: Null check and Duplication values check

```python
import pandas as pd

# Check data types
print(df.dtypes)

# Inspect unique values for categorical variables
print("SEX:", df['SEX'].unique())
print("EDUCATION:", df['EDUCATION'].unique())
print("MARRIAGE:", df['MARRIAGE'].unique())
print("PAY_0:", df['PAY_0'].unique())

# Encode categorical variables
# SEX: keep numeric
# SEX: keep numeric
df['SEX'] = pd.to_numeric(df['SEX'], errors='coerce')
df['SEX'] = df['SEX'].map({1:0, 2:1})   # optional binary encoding

# EDUCATION anomalies
df['EDUCATION'] = df['EDUCATION'].replace({0:4, 5:4, 6:4})

# MARRIAGE anomalies
df['MARRIAGE'] = df['MARRIAGE'].replace({0:3})

# Drop any rows where SEX is NaN (caused by header row slip)
df = df.dropna(subset=['SEX'])

# Repayment status (PAY_0 to PAY_6): already ordinal, but check ranges
for col in ['PAY_0','PAY_2','PAY_3','PAY_4','PAY_5','PAY_6']:
    print(col, df[col].unique())
```

```
ID                  object
LIMIT_BAL           object
SEX                 object
EDUCATION           object
MARRIAGE            object
AGE                 object
PAY_0               object
PAY_2               object
PAY_3               object
PAY_4               object
PAY_5               object
PAY_6               object
BILL_AMT1           object
BILL_AMT2           object
BILL_AMT3           object
BILL_AMT4           object
BILL_AMT5           object
BILL_AMT6           object
PAY_AMT1            object
PAY_AMT2            object
PAY_AMT3            object
PAY_AMT4            object
PAY_AMT5            object
PAY_AMT6            object
DEFAULT_NEXT_MONTH  object
dtype: object
SEX: ['SEX' '2' '1']
EDUCATION: ['EDUCATION' '2' '1' '3' '5' '4' '6' '0']
MARRIAGE: ['MARRIAGE' '1' '2' '3' '0']
PAY_0: ['PAY_0' '2' '-1' '0' '-2' '1' '3' '4' '8' '7' '5' '6']
PAY_0 ['2' '-1' '0' '-2' '1' '3' '4' '8' '7' '5' '6']
PAY_2 ['2' '0' '-1' '-2' '3' '5' '7' '4' '1' '6' '8']
PAY_3 ['-1' '0' '2' '-2' '3' '4' '6' '7' '1' '5' '8']
PAY_4 ['-1' '0' '-2' '2' '3' '4' '5' '7' '6' '1' '8']
PAY_5 ['-2' '0' '-1' '2' '3' '5' '4' '7' '8' '6']
PAY_6 ['-2' '2' '0' '-1' '3' '6' '4' '7' '8' '5']
```

- Confirms data types.
- Shows unique values for categorical variables.
- Cleans anomalies in EDUCATION (values 0, 5, 6 → "Others") and MARRIAGE (value 0 → "Others").
- Maps SEX into readable labels.
- Validates repayment status ranges (–2 to 9).

Fig 5: Unique value check



```
num_cols = ['LIMIT_BAL','BILL_AMT1','BILL_AMT2','BILL_AMT3',
            'BILL_AMT4','BILL_AMT5','BILL_AMT6',
            'PAY_AMT1','PAY_AMT2','PAY_AMT3',
            'PAY_AMT4','PAY_AMT5','PAY_AMT6']

# Convert to numeric, invalid entries become NaN
df[num_cols] = df[num_cols].apply(pd.to_numeric, errors='coerce')

# Drop rows with NaN in these columns
df = df.dropna(subset=num_cols)

# Now safe to compute outliers
from scipy import stats
outlier_counts = {col: (abs(stats.zscore(df[col])) > 3).sum() for col in
num_cols}

print("Outlier counts per column:")
print(outlier_counts)
```

```
Outlier counts per column:
{'LIMIT_BAL': np.int64(130), 'BILL_AMT1': np.int64(686), 'BILL_AMT2': np.int64(670), 'BILL_AMT3': np.int64(661), 'BILL_AMT4': np.int64(680), 'BILL_AMT5': np.int64(651), 'BILL_AMT6': np.int64(653), 'PAY_AMT1': np.int64(390), 'PAY_AMT2': np.int64(387), 'PAY_AMT3': np.int64(362), 'PAY_AMT
```

Fig 6: Checking for outliers

```
# Handle Outliers
# Scale numeric variables
from sklearn.preprocessing import RobustScaler

# Continuous variables
numeric_cols = ['LIMIT_BAL','BILL_AMT1','BILL_AMT2','BILL_AMT3',
                'BILL_AMT4','BILL_AMT5','BILL_AMT6',
                'PAY_AMT1','PAY_AMT2','PAY_AMT3',
                'PAY_AMT4','PAY_AMT5','PAY_AMT6']

# Initialize RobustScaler (less sensitive to outliers)
scaler = RobustScaler()

# Fit and transform
df[numeric_cols] = scaler.fit_transform(df[numeric_cols])

# Quick check: scaled summary
print(df[numeric_cols].describe())
```

```
       LIMIT_BAL     BILL_AMT1     BILL_AMT2     BILL_AMT3     BILL_AMT4  \
count  30000.000000  30000.000000  30000.000000  30000.000000  30000.000000
mean       0.144654      0.453972      0.458512      0.468267      0.463996
std        0.682882      1.159031      1.166372      1.206108      1.232920
min       -0.684211     -2.958521     -1.490901     -3.084472     -3.623126
25%       -0.473684     -0.296271     -0.298505     -0.303004     -0.320535
50%        0.000000      0.000000      0.000000      0.000000      0.000000
75%        0.526316      0.703729      0.701495      0.696996      0.679465
max        4.526316     14.829154     15.776915     28.592059     16.721858

       BILL_AMT5     BILL_AMT6      PAY_AMT1      PAY_AMT2      PAY_AMT3  \
count  30000.000000  30000.000000  30000.000000  30000.000000  30000.000000
mean       0.458560      0.454730      0.889561      0.938844      0.832486
std        1.255426      1.242205      4.134618      5.529367      4.278727
min       -2.053348     -7.439659     -0.524214     -0.482121     -0.437424
25%       -0.337443     -0.329876     -0.274588     -0.282217     -0.342649
50%        0.000000      0.000000      0.000000      0.000000      0.000000
75%        0.662557      0.670124      0.725412      0.717783      0.657351
max       18.771700     19.702726    217.536695    403.707703    217.312272

       PAY_AMT4      PAY_AMT5      PAY_AMT6
count  30000.000000  30000.000000  30000.000000
mean       0.894768      0.873085      0.957049
std        4.214449      4.042949      4.579166
min       -0.403524     -0.396930     -0.386374
25%       -0.323895     -0.330114     -0.356044
50%        0.000000      0.000000      0.000000
75%        0.676105      0.669886      0.643956
max      166.655458    112.471289    135.788782
```

Fig 7: Handling outliers

```python
#Feature Engineering

import numpy as np

# Safe ratio calculation: avoid division by zero
for i in range(1, 7):
    # Payment-to-Bill Ratio
    df[f'PAY_RATIO{i}'] = np.where(df[f'BILL_AMT{i}'] > 1e-6,   # tiny threshold
                                   df[f'PAY_AMT{i}'] / df[f'BILL_AMT{i}'],
                                   0)
    # Clip extreme ratios to a reasonable bound (e.g., 0-5)
    df[f'PAY_RATIO{i}'] = df[f'PAY_RATIO{i}'].clip(0, 5)

    # Utilization Ratio
    df[f'UTIL_RATIO{i}'] = np.where(df['LIMIT_BAL'] > 1e-6,
                                    df[f'BILL_AMT{i}'] / df['LIMIT_BAL'],
                                    0)
    # Clip utilization to 0-5 (500% usage cap)
    df[f'UTIL_RATIO{i}'] = df[f'UTIL_RATIO{i}'].clip(0, 5)

# After feature engineering, replace any residual inf/-inf with NaN and impute
df = df.replace([np.inf, -np.inf], np.nan)
df = df.fillna(0)   # or use median imputation if preferred


# Quick check: engineered features
print(df[['TOTAL_BILL','TOTAL_PAY','AVG_PAY_RATIO','BILL_TREND','PAY_TREND',
          'MISSED_PAYMENTS','OVERPAYMENTS']].head())
```

```
   TOTAL_BILL  TOTAL_PAY  AVG_PAY_RATIO  BILL_TREND  PAY_TREND  \
1   -2.019718  -2.465241       0.000000   -0.065379   0.137840
2   -1.824928  -1.363412       0.000000    0.022016   0.653005
3   -0.310574   0.172878      -0.224331   -0.139684   1.046821
4    2.000102  -0.518820      -0.326016   -0.127109  -0.103829
5   -0.096637  11.879900       8.022933    0.259622  -0.186513

   MISSED_PAYMENTS  OVERPAYMENTS
1                0             0
2                0             4
3                0             1
4                0             0
5                0             4
```

Fig 8: Feature Engineering

```python
#Class Imbalance Check
import matplotlib.pyplot as plt

# Target distribution
class_counts = df['CreditRisk'].value_counts()
print(class_counts)

# Percentage distribution
print(class_counts / len(df))

# Quick bar plot
class_counts.plot(kind='bar', title='Class Distribution')
plt.show()
```

```
CreditRisk
0    23364
1     6636
Name: count, dtype: int64
CreditRisk
0    0.7788
1    0.2212
Name: count, dtype: float64
```

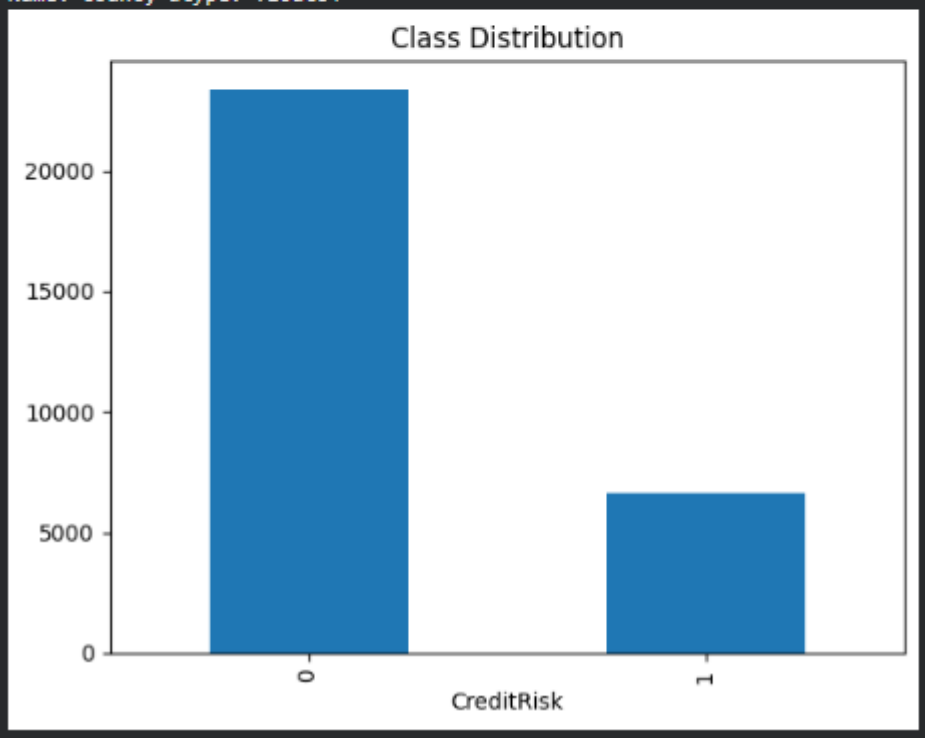


Fig 9: Class Imbalance check

```python
# Step 1: Clean target column
df["CreditRisk"] = pd.to_numeric(df["CreditRisk"], errors="coerce")

df = df.dropna(subset=["CreditRisk"])
df["CreditRisk"] = df["CreditRisk"].astype(int)

# Step 2: Rename to meaningful name
#df = df.rename(columns={"DEFAULT_NEXT_MONTH": "CreditRisk"})


print(df["CreditRisk"].value_counts())
```

```
CreditRisk
0    23364
1     6636
Name: count, dtype: int64
```

Fig 10: Cleaning target column

```python
import numpy as np
import pandas as pd

# Force all columns to numeric (anything non-numeric becomes NaN)
df_numeric = df.apply(pd.to_numeric, errors="coerce")

# Check for infinities
inf_mask = np.isinf(df_numeric.to_numpy())
print("Total infinities:", inf_mask.sum())
#print("Infinities per column:\n", pd.Series(inf_mask.sum(axis=0),
index=df_numeric.columns))

# Check for NaNs
nan_mask = np.isnan(df_numeric.to_numpy())
print("Total NaNs:", nan_mask.sum())
#print("NaNs per column:\n", pd.Series(nan_mask.sum(axis=0), index=df_numeric.
columns))

# Identify problematic rows
bad_rows = np.where(inf_mask.any(axis=1) | nan_mask.any(axis=1))[0]
print("Rows with issues:", bad_rows[:20])  # show first 20

```

```
Total infinities: 0
Total NaNs: 0
Rows with issues: []
```

Fig 11: Checking for NaN or Infinity values

```python
# Step 5: Train/Test split
from sklearn.model_selection import train_test_split

# Target variable: DEFAULT_NEXT_MONTH (already binary: 0 = Non-default, 1 = Default)
X = df.drop(columns=["ID","CreditRisk"])   # drop ID and target
y = df["CreditRisk"]

# Check distribution
print("Class distribution:\n", y.value_counts())

# Train/Test split (stratified to preserve class distribution ~22% default)
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42, stratify=y
)
```

```
Class distribution:
 CreditRisk
0    23364
1     6636
Name: count, dtype: int64
```

Fig 12: Test Train Split

Base Model

```python
# Step 6: Baseline Logistic Regression
model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
```

- Reasoning: Logistic Regression is a baseline, interpretable, widely used in credit scoring.
- Assumptions: Linear relationship between predictors and log-odds of default.
- Limitations: Sensitive to multicollinearity, may underperform on nonlinear patterns.

```python
# Step 7: Evaluation
from sklearn.metrics import accuracy_score

print("Confusion Matrix:\n", confusion_matrix(y_test, y_pred))
print("\nClassification Report:\n", classification_report(y_test, y_pred))
acc = accuracy_score(y_test, y_pred)
print("\nAccuracy:", round(acc, 2))
```

```
Confusion Matrix:
 [[6799  210]
 [1522  469]]

Classification Report:
              precision    recall  f1-score   support

           0       0.82      0.97      0.89      7009
           1       0.69      0.24      0.35      1991

    accuracy                           0.81      9000
   macro avg       0.75      0.60      0.62      9000
weighted avg       0.79      0.81      0.77      9000


Accuracy: 0.81
```

Fig 13: Logistic regression, confusion matrix and Classification Report